

Ethan Lv

Beijing, China | thulvcc2017@gmail.com | 186 1078 3566

Summary

LLM specialist with 4 years of end-to-end experience spanning data strategy, pre-training, post-training alignment (RLHF / DPO / GRPO / OPD), and AI Agent infrastructure. Delivered 70B-scale model training with 10T+ token data pipelines; published 6 papers at NeurIPS / EMNLP / ACL, including C-Eval (NeurIPS 2023), a widely-adopted Chinese LLM benchmark. Currently focused on Agent infrastructure, online preference optimization, and continual model improvement at WeChat scale (100M+ DAU).

Experience

Tech Lead, WeChat AI Assistant, Tencent Technology (Beijing) – Beijing, China May 2025 – present

- Architected the WeChat AI Assistant Agent Harness from scratch — designing and implementing tool-calling orchestration, multi-step planning and reasoning, context memory management, and multi-layer safety guardrails for a system serving hundreds of millions of daily active users
- Led Agent-specific post-training, developing systematic training recipes across instruction following, tool-use alignment, multi-turn consistency, and safety refusal boundaries; introduced Online Preference Distillation (OPD) to enable continuous preference alignment post-deployment, leveraging Iterative DPO, GRPO, and Constitutional AI
- Built a multi-dimensional Agent evaluation matrix covering single-step tool accuracy, multi-step task completion rate, hallucination rate, safety boundary adherence, and long-horizon consistency, establishing a data flywheel for joint model–framework optimization
- Drove deep integration of Agent capabilities with the WeChat ecosystem, designing a standardized tool-access protocol that unified tool schemas across Mini Programs, Official Accounts, WeChat Pay, and Search, significantly improving end-to-end task success rates

LLM Algorithm Engineer, Shenyang Technology (Beijing) – Beijing, China

Mar 2022 – Apr 2025

- Led data strategy and training pipeline construction for general-purpose foundation models: designed multi-stage filtering pipelines for pre-training and instruction-tuning data, combining heuristic rules, perplexity distribution analysis, and model-based quality annotation for fine-grained classification and deduplication, ultimately producing 10T high-quality pre-training tokens and millions of premium SFT samples
- Built a comprehensive evaluation framework aligned with training data taxonomy, integrating mainstream benchmarks with proprietary test suites (including temporally relevant and rigorously leaked evaluation data), establishing a closed-loop of data classification → evaluation alignment → capability diagnosis
- Designed a Scaling Law experiment matrix grounded in fine-grained data labeling, systematically validating the impact of data mixture ratios, hyperparameters, and curriculum strategies on loss convergence and downstream performance across small-scale models (100M–several B), identifying optimal pre-training recipes that generalized successfully to the full 70B model series
- Led multi-dimensional capability improvement initiatives: executed phased long-context extension (8K → 32K → 128K) via RoPE base-frequency tuning, data-mixture optimization, and progressive training; simultaneously drove specialized enhancements in STEM reasoning and code generation, with results approaching state-of-the-art models of comparable scale
- For the financial vertical domain, constructed 100B high-quality CPT data from 40T+ raw multi-source corpora; designed and implemented a document-level curriculum learning scheme that organized training data by difficulty and inter-document semantic similarity, combined with dynamic hyperparameter scheduling to effectively mitigate catastrophic forgetting
- Delivered long-text summarization and intelligent Q&A product features atop internal LLMs by decomposing tasks into evaluable sub-modules (extraction–compression–rewriting–verification), performing independent data synthesis, quality filtering, and targeted training per module, forming a reusable data flywheel
- Applied DPO training on auto-constructed preference pairs to substantially reduce hallucination and repetitive generation; implemented GRPO training with citation recall as the reward signal, achieving a step-change improvement in generated citation accuracy

Education

Tsinghua University, M.S. in Computer Science and Technology – Beijing, China

Aug 2021 – July 2024

- Research focus: Natural Language Processing and Large Language Models

Tsinghua University, B.S. in Computer Science and Technology – Beijing, China

Aug 2017 – July 2021

Publications

GATEAU: Selecting Influential Samples for Long Context Alignment

Nov 2025

Shuzheng Si, Haozhe Zhao, Gang Chen, Yunshui Li, Kangyang Luo, *Ethan Lv*, Kaikai An, Fanchao Qi, Baobao Chang, Maosong Sun

[10.18653/v1/2025.emnlp-main.375](https://arxiv.org/abs/10.18653/v1/2025.emnlp-main.375) (EMNLP 2025)

Document Segmentation Matters for Retrieval-Augmented Generation

July 2025

Zhitong Wang, Cheng Gao, Chaojun Xiao, Yufei Huang, Shuzheng Si, Kangyang Luo, Yuzhuo Bai, Wenhao Li, Tangjian Duan, *Ethan Lv*, Guoshan Lu, Gang Chen, Fanchao Qi, Maosong Sun

[10.18653/v1/2025.findings-acl.422](https://arxiv.org/abs/10.18653/v1/2025.findings-acl.422) (ACL 2025 Findings)

HyperLoRA: Efficient Cross-task Generalization via Constrained Low-Rank Adapters Generation

Nov 2024

Ethan Lv, Lei Li, Shitou Zhang, Gang Chen, Fanchao Qi, Ningyu Zhang, Hai-Tao Zheng

[10.18653/v1/2024.findings-emnlp.956](https://arxiv.org/abs/10.18653/v1/2024.findings-emnlp.956) (EMNLP 2024 Findings)

C-Eval: A Multi-Level Multi-Discipline Chinese Evaluation Suite for Foundation Models

Dec 2023

Yuzhen Huang, Yuzhuo Bai, Zhihao Zhu, Junlei Zhang, Jinghan Zhang, Tangjun Su, Junteng Liu, *Ethan Lv*, Yikai Zhang, Jiayi Lei, Fanchao Qi, Yao Fu, Maosong Sun, Junxian He
[10.48550/arXiv.2305.08322](https://arxiv.org/abs/10.48550/arXiv.2305.08322) (NeurIPS 2023 (Datasets & Benchmarks))

Sememe Prediction for BabelNet Synsets using Multilingual and Multimodal Information

May 2022

Fanchao Qi, *Ethan Lv*, Zhiyuan Liu, Xiaojun Meng, Maosong Sun, Haitao Zheng
[10.48550/arXiv.2203.07426](https://arxiv.org/abs/10.48550/arXiv.2203.07426) (ACL 2022 Findings)

Temporal Cross-Effects in Knowledge Tracing

Mar 2021

Chenyang Wang, Weizhi Ma, Min Zhang, *Ethan Lv*, Fengyuan Wan, Huijie Lin, Taoran Tang, Yiqun Liu, Shaoping Ma
[10.1145/3437963.3441802](https://arxiv.org/abs/10.1145/3437963.3441802) (WSDM 2021)

Skills

LLM Training & Alignment: Scaling Law experiments & pre-training strategy, SFT data strategy, Iterative DPO / GRPO / OPD / Constitutional AI, CPT with anti-forgetting techniques

AI Agent Systems: Agent framework architecture, function calling & tool orchestration, multi-step planning & reasoning chains, memory & context management, safety guardrails

Data Engineering: Large-scale corpus filtering & deduplication (10T+), data synthesis & augmentation, automated annotation & classification, multi-dimensional data evaluation

Model Evaluation: Benchmark construction (C-Eval), multi-dimensional capability diagnosis, fair evaluation design, leaked evaluation data management

Specialized Capabilities: Long-context extension (128K), RAG system design & optimization, STEM / code / reasoning enhancement, catastrophic forgetting mitigation

Tech Stack: Python, PyTorch, DeepSpeed, Transformers, distributed training (FSDP / Megatron), vector databases & retrieval systems